

A283 – Thermodynamics in FFGFT: a comprehensive overview

Johann Pascher

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Abstract. Thermodynamics in FFGFT is anchored on five levels – not as a postulate, but as a consequence of geometry. This document consolidates the distributed individual results, organizes them into a coherent argumentative chain, and identifies where the levels interlock and where they remain separate. A concluding section acknowledges that both framework descriptions – expanding space and fractal refinement – are models that interpret the same observations in structurally opposing ways.

Layer markers. [SET] declared starting point — [B] algebraically/mathematically derived from declared foundations — [K] numerically calculated from corpus constants — [Q] external primary source — [S] plausibility sketch — [H] open research question.

.1 Why a comprehensive document

The thermodynamic content of FFGFT has been developed across several documents: Doc. 157 addresses the arrow of time, A271–A282 cover Landauer accounting, and Doc. 312/313 treat cosmological thermodynamics. Each document is self-contained; none provides an overview of how the parts connect.

[B] Three questions remain unanswered without a comprehensive document: (1) Where do the levels touch, and where do they remain separate? (2) Which statements depend on the ξ -scheme, and which are independent of it? (3) What is the epistemic status of each statement? This document answers these questions and contains no new substantive claims.

.2 Level 1 – The arrow of time is geometric (Doc. 157)

The standard problem

All fundamental equations of physics are time-reversal invariant. The second law ($\Delta S \geq 0$) is statistical in nature and shifts the problem to the unexplained low initial entropy (Penrose: $\sim 10^{-10^{123}}$).

The FFGFT answer

[B] Three geometric arguments interlock (Doc. 157):

Argument 1: Algebraic asymmetry

$\tilde{T} \cdot m = 1$ enforces $\tilde{T} > 0$ because mass is definitively positive. Time reversal would require $m \rightarrow -m$ – physically impossible.

Argument 2: Topological winding direction

The vacuum field winds with $\dot{\theta} = m = 1/\tilde{T}$ along the T^4 . This chirality is topologically protected.

Argument 3: Fractal dimensional asymmetry

$D_f = 3 - \xi_0 \approx 2.9999$ breaks exact mirror symmetry; path integrals for forward and backward directions differ by $O(\xi_0)$.

Consequence

[B] The logic reverses: $\Delta S \geq 0$ is in FFGFT a *consequence* of geometry, not a postulate. The "Past Hypothesis" becomes unnecessary – the low initial entropy is geometrically enforced, not an initial condition.

.3 Level 2 – Landauer as phase-space accounting (A271)

The phase-space formulation

[SET] A physical system occupies a region in phase space of size W (measured in h^f). The region exists independently of how one labels it.

[B] A mapping that reduces multiple input regions $\{A, B\}$ to a single target region Z reduces the volume: $\Delta S_{\text{system}} = -k_B \ln(W_{A+B}/W_Z) \leq 0$. Liouville's theorem forbids volume shrinkage in a closed system. A heat bath must absorb the volume:

$$Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

with $\ln 2$ as the binary special case.

Three separations

[B] (1) *Bit $\neq$ physical region*. Shannon and Boltzmann entropy are identical only under uniform distribution over physical microstates. (2) *Content neutrality*. Thermodynamics counts regions, does not read them. Bit "true" and bit "false" cost identically. (3) *Three levels separated*. Construction / state transition / description. Landauer costs occur at levels 1 and 2, not at 3.

.4 Level 3 – Phase-space accounting universal (A280–A282)

[B] A280: The region comes first. A glass into the ocean – the volume does not disappear, it distributes over $1.4 \times 10^9 \text{ km}^3$. The price: the bath must absorb the volume, heat flows. The counting basis is distinguishability (orthogonality), not energy; absolute fundamental discretization: h^f .

[B] A281: Landauer's own restriction – logical irreversibility is a description property, not a physical one. Sufficient for heat is solely the non-bijectivity of the region mapping. Bennett's result: Every logically irreversible function can be executed as a sequence of bijective mappings; costs stick to *forgetting*, not to the operation.

[B] A282: The token computer as a limiting case. Accounting holds exactly, Landauer's bound does *not* apply there (engineering costs dominate). The limit to the Brownian computing sphere continuously connects to Landauer's domain of validity.

[H] Open question: Minimal coherence cost for quantum coherence independent of error correction (A271, § 7.4).

.5 Level 4 – Cosmological thermodynamics (Doc. 312/313)

Thermal history as geometric condition

[K] Doc. 313: Three phases on the observable half-cycle ($\tau_c/2 \approx 46$ Gyr):

Phase	Range	θ/π	Character
Radiation domination	$z > 3000$	> 0.90	$m \rightarrow m_{\min}$, clocks slowest
Transition	$z \approx 3000\text{--}1090$	$0.90\text{--}0.979$	Recombination, scattering surface
Matter domination	$z < 3000$	< 0.90	$m(\theta)$ grows, structure formation

The five-step chain

[K] Complete derivation of Ω_m^* from ξ_0 alone:

Step	Result
(1) $H_0 = (\pi/2) c \xi_0^{10} / \bar{\lambda}_e$	$h = 0.6682$
(2) $k_B T_0 = (16/9) \xi_0$ (Doc. 061)	$T_0 = 2.751$ K
(3) $\Omega_r \propto T_0^4 / H_0^2$	9.643×10^{-5}
(4) $K_{\text{frak}} = 1 - 100 \xi_0$ (A040)	0.98667
(5) $\int_0^\infty dz/E(z) = \pi/K_{\text{frak}}$	$\Omega_m^* = 0.3136$

Planck: $\Omega_m = 0.315 \pm 0.007$ (-0.19σ). No step contains a parameter fitted to Ω_m . Radiation ($\Omega_r \propto T_0^4 / H_0^2$) is the link between the particle and cosmic sectors.

Overdetermination and open core

[K] T_0 is determined via two independent paths: Particle sector (Path 1: $k_B T_0 = (16/9) \xi_0$) and cosmic sector (Path 2: antipode condition). Both yield Ω_m agreeing to 0.1 %.

[K] Open (D(ii)): Coarse-grained entropy on the return half-cycle. Poincaré discrepancy: $\sim 10^{104}$ dynamical times versus $\tau_c \approx 92$ Gyr. Periodicity is a boundary condition, not a consequence of dynamics. Three cases (A050/Doc. 295): Case A ($d = 0, 1$ revolution), Case B (ξ frozen, 75 revolutions), Case C (ξ runs, logarithmic spiral – removes entropy requirement, but costs τ_c).

Falsification points

[K] Low- ℓ cutoff: $\ell_{\min} = 2\pi D_{\text{LSS}}/L^* = 3.08$ – hits the quadrupole suppression unexplained since COBE (1992). Primary falsification test: $m_\tau = 1776.97$ MeV (Belle-II).

.6 Level 5 – What separates the levels

[B] Landauer accounting (Levels 2 and 3) does not touch the ξ -scheme. It holds in any system with Hamiltonian or unitary dynamics – independent of the torus geometry. Explicitly logged in A271: "Contact with the FFGFT ξ -scheme: none."

The arrow of time (Level 1), by contrast, is directly tied to $D_f = 3 - \xi_0$ and $\tilde{T} \cdot m = 1 - \text{FFGFT-specific}$. Cosmological thermodynamics (Level 4) depends entirely on ξ_0 , H_0 , and the torus geometry.

Level	Document	ξ -dependence	Status
Arrow of time	Doc. 157	direct	[B]/[S]
Landauer	A271	none	[B]
Universal accounting	A280–A282	none	[B]
Cosmol. thermodynamics	Doc. 312/313	direct	[K]
Information physics	Doc. 243–301	indirect	[S]/[B]

.7 Argument chain: from geometry to heat

[B] The five levels form a chain from FFGFT geometry to classical thermodynamics: (1) ξ_0 and T^4 fix the geometry. (2) $\tilde{T} \cdot m = 1$ and the torus winding define geometrically a direction of time. (3) In this directed time, there is more phase-space volume in the forward direction – $\Delta S \geq 0$ as geometric consequence. (4) Non-bijective phase-space mappings (Landauer) cost heat – independent of ξ_0 , in the same phase-space picture. (5) The thermal history (Ω_m^* , Λ^*) follows from ξ_0 directly.

Liouville's theorem and the second law are in this chain *consequences*, not postulates.

.8 Information thermodynamics in the IPI context (Doc. 243–301)

[S] A further group of documents connects FFGFT thermodynamics to the information physics community (IPI context): Doc. 251 (comparison with Vopson's information mass hypothesis), Doc. 247 (Landauer's principle in the context of information states), Doc. 290 (information question magnitude/phase), Doc. 301 (information reversal test). These documents are plausibility sketches or partial algebraic derivations; they open connections to current debates, without contributing new core results to FFGFT thermodynamics.

.9 Phases, radiation, and temperature in the time cycle (Doc. 313 in detail)

Gradient on a circle

[B] The observable history shows a clear thermal sequence: radiation domination, transition, matter domination. This gives the impression of a directed process with a beginning. The FFGFT answer (Doc. 313): The gradient is not an evolution *from* a beginning, but a map *along* the time cycle – like climate zones along a meridian circle.

Three phases on the half-cycle

[K] The observable history fills a half-cycle ($\tau_c/2 \approx 46$ Gyr). Cycle positions:

Event	z	θ/π	
		smooth	fractal
Matter-radiation equality	~ 3400	0.93	0.92
Scattering surface (recombination)	1090	0.992	0.979
Particle horizon	$z_{\max}$	1.012	0.9986

Phase 1 — Radiation domination (near the antipode)

In the mass-run reading: masses $m(\theta)$ are minimal, intrinsic times $\tilde{T} = 1/m$ maximal – clocks run slowest. The endpoint slope of the mass profile at the antipode:

$$\left| \frac{dm}{d\theta} \right|_{\text{antipode}} = \sqrt{\Omega_r} \approx 0.0098$$

No free parameter – the observed radiation density appears here as a *smoothness condition* of the periodic closure.

Phase 2 — Matter-radiation transition ($z \approx 3000$ to **1090**)

With the fractal path correction $K_{\text{frak}} = 1 - 100\xi_0$ (A040, independently derived), the particle horizon sits at 0.14 % from the antipode – improvement by factor 9 over the smooth calculation (1.2 %), without fitted parameter.

The path correction affects path/structure, shortens path/path. Invariance test: CMB peak positions remain unchanged (test passed without having been designed for it).

Phase 3 — Matter domination (present epoch)

Energy density $\propto (1+z)^3$. In the mass-run reading, masses grow with decreasing z ; clocks run faster; structure formation unfolds.

Temperature as a structural quantity

[K] In the standard model: $T(z) = T_0(1+z)$ – adiabatic back-extrapolation. In FFGFT: T_0 comes from the atomic scale (Doc. 061):

$$k_B T_0 = \frac{16}{9} \xi_0 \approx 2.370 \times 10^{-4} \text{ eV} \quad (\text{measurement: } 2.349 \times 10^{-4} \text{ eV, } +0.92 \%)$$

The CMB temperature is a *structural quantity* of the ξ -scheme, not the endpoint of a cooling history.

[K] Range limit (Doc. 313): Equilibria at other locations of the time cycle are *not* derivable from today by forward-extrapolation. Unaffected: the Ω_m^* -chain (runs via ξ_0 , not via adiabatic back-calculation).

Overdetermination of T_0

[K] Two independent paths: Path 1 (particle sector): $k_B T_0 = (16/9)\xi_0$, no cosmological input. Path 2 (cosmic sector): antipode condition, no particle physics input. Both yield Ω_m agreeing to 0.1 % (0.3136 vs. 0.3139). Leverage: $dT_0/T_0 / d\Omega_m/\Omega_m \approx -12$.

Lower bound and Hawking

[K] Energy quantum of the time cycle (Doc. 312): $E_{\min} = \hbar H_0 = 2.28 \times 10^{-52} \text{ J}$. Finite $z_{\max} = m_e/m_{\min} - 1 = 3.59 \times 10^{38}$. Singularity excluded: masses cannot run to zero.

[K] A KMS rule connects Gibbons–Hawking, Unruh, and Hawking: $T = \hbar/(k_B \tau)$ with τ as local time period. The compact time direction yields T_{GH} without horizon. The family ladder meets the cosmos at $r_s = R_H/2$ (exactly 0.500). Hawking evaporation for $M_\odot$: 56 orders of magnitude too slow for τ_c – sharpens the entropy obstacle D(ii).

Low- ℓ cutoff

[K] On a torus of side length L^* , modes with wavelength $> L^*$ are missing:

$$\ell_{\min} = \frac{2\pi D_{\text{LSS}}}{L^*} = 3.08$$

No fitted parameter. Hits the quadrupole suppression ($\ell = 2$, factor $\sim 1/6$), an unexplained finding in ΛCDM since COBE (1992). Order statement [K|P20]; exact mode summation on the torus remains pending.

Robust and non-robust statements

[K] Three classes (Doc. 313):

Class	Examples	Robustness
Counts	$ \text{Aut}(D4) = 1152$, kissing number 24	exact
Ratios on same level	α , m_μ/m_e , Koide Q	to 10^{-10}
With scale bridge	G , H_0 , absolute masses	only with anchor
Number and duration quantities	$\eta = n_B/n_\gamma$, window durations	not transferable

BBN observables (Y_p , D/H , ${}^7\text{Li}$): not a pure ratio on the same level – the same holds for ΛCDM .

.10 Mass formation, fractal refinement, and the appearance of expansion

What is missing at the antipode: metric sharpness

[K] All physical lengths that depend on mass scale with $1/m$: Compton wavelength $\bar{\lambda} = \hbar/(mc)$, Bohr radius $a_0 \propto 1/m_e$, intrinsic time $\tilde{T} = 1/m$. At the antipode ($m \rightarrow m_{\min}$), all run to their maximum. Without localizable massive objects, *metric sharpness* is missing.

[H] What metric structure precisely means at $m \rightarrow m_{\min}$ has not yet been elaborated in the corpus.

Photons: no freezing

[B] Photons carry no rest mass – $\tilde{T} \cdot m = 1$ does not apply to them directly. A photon at the antipode does not freeze; it propagates at c . CMB photons lose energy geometrically on their way to us (A265):

$$\frac{dE}{dx} = -\xi E \Rightarrow 1+z = e^{\xi x}$$

Achromatic: all wavelengths stretched equally. From our perspective, everything runs uniformly. Looking back is looking at a location on the time cycle, not looking into a past.

Expansion as fractal refinement inward

[B] In the standard model: space expands outward, scale factor $a(t)$ grows. In FFGFT, there is no expanding space. With increasing $m(\theta)$, all characteristic lengths ($\propto 1/m$) become smaller. Matter becomes more structured, more finely resolved. This is **refinement inward**, not growth outward.

[B] $K_{\text{frak}} = 1 - 100\xi_0$ (A040): dimension deficit $D_f = 3 - \xi_0$, locally $\sim 0.003\%$, cumulative over 17 orders of magnitude ~ 100 .

[B] Recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (A050), fixed point: zero. Rolled up: winding closes after 75 revolutions (τ_c -cycle). Unrolled: logarithmic spiral $\rho = 75 e^{\beta/2\pi}$ – what the standard model calls expansion.

[K] Both descriptions are observationally equivalent (A265). Rulers and clocks change in the same ratio; no local observer notices the scaling. What we see as expansion is the difference between our present masses and the redshifted light from the antipode.

[S] Image: Mandelbrot zoom. Zooming inward reveals ever finer structures, no outward growth. Antipode = unzoomed view (minimal differentiation, pure radiation). Present position = highly zoomed view (sharp atomic and stellar structures). The standard model turns the image around – a coordinate choice, not a physical statement.

.11 Minimal length and fractal refinement: an apparent contradiction

[S] At first glance, there appears to be a tension: On the one hand, FFGFT speaks of a *minimal length* at the antipode ($\bar{\lambda} = \hbar/(m_{\text{min}}c)$), on the other hand, of a *fractal refinement inward* along the time cycle. Is this a contradiction?

The answer is **no** – the key lies in the *reference-frame logic* of the theory.

What does "minimal length" mean?

The "minimal length" in FFGFT is *not* an absolute Planck length, but a **mass-dependent quantity**:

$$\bar{\lambda} = \frac{\hbar}{mc}, \quad a_0 \propto \frac{1}{m_e}, \quad \tilde{T} = \frac{1}{m}.$$

At the antipode ($m \rightarrow m_{\text{min}}$), these lengths become *maximal*. There, massive structures are least sharply resolved – a regime of *minimal metric sharpness*, dominated by radiation. The "minimal length" is thus actually the *largest characteristic length* of the system.

What does "fractal refinement inward" mean?

With increasing mass $m(\theta)$ along the time cycle, all characteristic lengths become *shorter*:

$$\bar{\lambda} \propto \frac{1}{m(\theta)}, \quad a_0 \propto \frac{1}{m_e(\theta)}.$$

This means: matter becomes *more differentiated, more structured, more finely resolved* – like a **zoom inward**. Space itself does not expand; the *scale of objects* changes.

Why is this not a contradiction?

The apparent tension dissolves once one distinguishes two *different perspectives*:

Perspective	Meaning in FFGFT
Minimal length (at the antipode)	Maximal Compton wavelength, minimal structuring, radiation-dominated regime
Fractal refinement inward	Increasing differentiation, growing masses, shorter wavelengths, present matter domination

The **direction** is unambiguous:

$$\text{Antipode } (m_{\min}, \bar{\lambda}_{\max}) \rightarrow \text{present position } (m_0, \bar{\lambda}_0)$$

Here, *structural resolution increases* (fractal inward), while *Compton lengths decrease*. This is not a contradiction, but a **complementary description** of the same process.

Comparison with the standard model

The standard model says: "The space expands, wavelengths are stretched."

FFGFT says: "Masses grow, wavelengths shrink, structure becomes finer."

Both descriptions are **observationally equivalent** (A265), because *clocks and rulers change together* – no local observer can measure the scaling.

Conclusion

The concepts of "minimal length" and "fractal refinement inward" are *not* a contradiction, but two sides of the same geometric coin:

- **Length perspective:** from large (antipode) to small (today)
- **Structure perspective:** from coarse (antipode) to fine (today)

Both follow from the same mass profile $m(\theta)$ and are consistently connected in FFGFT.

[S] This clarification makes visible the internal consistency of FFGFT geometry and resolves a common point of misunderstanding.

.12 Open points

[H] Three questions remain open: (1) Minimal coherence cost for quantum coherence independent of error correction (A271, § 7.4). (2) Continuous entropy flow as a function of clock cycle, bit width, and logical irreversibility (A271, § 8). (3) Metric structure in the limit $m \rightarrow m_{\min}$ – whether a regime of pure field geometry without localizable particle modes exists.

.13 Layer status summary

Statement	Status
$\tilde{T} > 0$ enforces geometric arrow of time	[B]
Torus winding: topologically protected direction of time	[B]

Statement	Status
$D_f = 3 - \xi_0$ breaks time-reversal symmetry	[B]
$\Delta S \geq 0$ as geometric consequence	[B]
Landauer: $Q \geq k_B T \ln 2$ from Liouville	[B]
Content neutrality of thermodynamics	[B]
Bennett's reversible logic	[B]
$\Omega_m^* = 0.3136$ from ξ_0 -chain	[K]
$\Lambda^* = 1/R_H^2$ geometric carrier	[K]
Overdetermination T_0 : two sectors to 0.1 %	[K]
Singularity excluded by $\hbar H_0$	[K]
$\ell_{\min} = 3.08$ (Low- ℓ cutoff)	[K P20]
Slowest clocks at the antipode	[K]
Compton wavelength, Bohr radius $\propto 1/m$	[B]
Photons achronically redshifted: $1 + z = e^{\xi x}$	[B]
Expansion = fractal refinement inward	[B]
Recursion: fixed point zero, spiral in unrolled case	[B]
Observational equivalence with Λ CDM	[K]
Landauer does not touch ξ -scheme	[B]
Minimal length = maximal Compton wavelength	[S]
Fractal refinement = increasing structure	[S]
No contradiction – two sides of same geometry	[S]
Coarse-grained closure D(ii): open	open
$m = 0$ structurally excluded: no empty state (A060)	[B]
Double cutoff: $L_0 \Rightarrow E_{\max}, L^* \Rightarrow E_{\min}$	[K]
$m_{\min} = \hbar H_0 / c^2$ from IR cutoff (Doc. 312/313)	[K]
Local field modes at thinnest point resolvable?	[H]
Past Hypothesis as geometric necessity	[S]
Mandelbrot zoom as image	[S]

Balance: 13× [B] · 8× [K] · 1× [K|P20] · 5× [S] · 1× [H] · 1× open.

.14 Concluding remark: Two opposing models

[S] An honest acknowledgment belongs at the end.

For us – as observers with our present masses and clocks – our perspective naturally counts. We think in light paths. We measure redshifts. We experience time as running forward. This is real in the sense that it is what we measure, and our instruments are calibrated to it.

The standard model (Λ CDM) takes exactly this perspective as its starting point: expanding space, Big Bang as beginning, redshift as space stretching, CMB as snapshot of an early epoch. It is a model that agrees excellently with observations.

FFGFT proposes a different coordinate choice: no expanding space, no Big Bang as an event, redshift as geometric energy loss, CMB as a view of a location on the time cycle. This description also agrees with the standard tests (A265, cosmological degeneracy).

[B] What distinguishes the two models:

- Λ CDM generalizes from the observer perspective *outward*: expanding space.
- FFGFT generalizes from geometry *inward*: compact torus, fractal refinement.

Both are models. Both leave direct observation behind and add an interpretative layer.

[S] What remains: two structurally opposing ways of describing the same thing. In one, everything expands outward; in the other, everything refines inward. In one, time has a

beginning; in the other, it is a closed cycle. In one, the CMB is a window into the past; in the other, a window onto a location.

This is not a flaw – it is an open question, formulated precisely enough to be decidable. The falsification tests (m_r , $\ell_{\min}$, Ω_m^* -chain) will show whether the geometric inward perspective of FFGFT carries more than the expanding outward perspective – or less.

.15 References

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.16 Use of AI tools

This document was created with the assistance of AI-based tools. Questions, foundational decisions, and substantive choices originate with the author; the tools assist in formulation and organization. Responsibility for content and errors lies entirely with the author. Full wording in A010.